

Customer Shopping Behaviour Analysis

An End-to-End Data Analytics Study Using Python, SQL & Power BI

Prepared for Retail Customer Behaviour Analytics Project

Dataset: 3,900 Customer Transactions | 18 Attributes | 50 U.S. Locations

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1 Executive Summary

This report presents a comprehensive analysis of customer shopping behaviour based on a transactional dataset of 3,900 individual purchases spanning 18 customer and product attributes, including demographics, purchase amount, product category, review ratings, subscription status, discount usage, and shipping preferences. The objective of the analysis is to uncover actionable insights into revenue drivers, customer segmentation, product performance, and purchasing patterns that can inform merchandising, marketing, and retention strategy.

The analysis was conducted through a three-stage pipeline: (1) data cleaning and feature engineering in Python using pandas, (2) structured business-question analysis using SQL against a PostgreSQL database, and (3) interactive dashboard visualisation built in Microsoft Power BI. This document consolidates the findings from all three stages into a single, decision-ready report.

\$233,081 TOTAL REVENUE	3,900 TRANSACTIONS	\$59.76 AVG. ORDER VALUE	3.75 / 5 AVG. RATING
4 PRODUCT CATEGORIES	25 UNIQUE PRODUCTS	43.0% DISCOUNT USAGE	27.0% SUBSCRIBERS

Key Findings at a Glance

- Male customers generated 68% of transaction volume and \$157,890 in revenue (67.7% of total), more than double the revenue contributed by female customers (\$75,191).
- Clothing is the leading product category, generating \$104,264 in revenue (44.7% of total), followed by Accessories (\$74,200).
- Express shipping customers spend marginally more on average (\$60.48) than Standard shipping customers (\$58.46).
- 839 customers used a discount and still spent at or above the \$59.76 average purchase amount, showing discounts do not universally suppress order value.
- An overwhelming 79.9% of customers fall into the “Loyal” segment (more than 10 previous purchases), indicating a highly repeat-driven customer base.
- Subscribed customers do not out-spend non-subscribers on a per-order basis (\$59.49 vs. \$59.87 average); subscription status is not currently a strong revenue differentiator.
- Young Adult customers are the single largest revenue-contributing age group at \$62,143, narrowly ahead of Middle-Aged customers.

2 Introduction & Project Objectives

Understanding how customers shop - what they buy, how much they spend, and which factors influence loyalty and repeat purchasing - is central to building an effective retail strategy. This project was undertaken to analyse a real-world-style customer shopping behaviour dataset in order to answer specific, business-relevant questions about revenue drivers, product performance, and customer segmentation.

Project Objectives

- Clean and prepare raw transactional data for reliable analysis.
- Answer ten targeted business questions using SQL against a relational database.
- Engineer derived features (age groups, purchase frequency in days, customer segments) to enable deeper analysis.
- Visualise key metrics through an interactive Power BI dashboard for ongoing monitoring.
- Synthesise all findings into a single professional report with clear, actionable recommendations.

Scope of the Dataset

The dataset comprises 3,900 customer transaction records, each representing one purchase event, with 18 attributes covering customer demographics (age, gender, location), product details (item purchased, category, size, colour, season), transaction details (purchase amount, discount applied, promo code used, shipping type, payment method), and behavioural indicators (review rating, subscription status, previous purchases, frequency of purchases).

3 Methodology & Tools

The analysis followed a structured, three-stage analytics workflow, moving from raw data to business insight to visual storytelling. Each stage used a different tool suited to its purpose, mirroring standard industry practice for end-to-end analytics projects.

3.1 Data Preparation - Python (pandas)

Data cleaning and feature engineering were performed in a Jupyter notebook using the pandas library. The following steps were applied to the raw CSV export:

- Loaded the raw dataset (3,900 rows × 18 columns) and profiled it using `.info()`, `.describe()`, and `.isnull().sum()` to identify structural issues and missing values.
- Imputed missing Review Rating values using the median rating within each product Category, preserving category-level rating tendencies rather than applying a single global median.
- Standardised all column headers to lowercase, snake_case formatting for consistency and downstream SQL compatibility.
- Renamed the ambiguous “purchase_amount_(usd)” column to “purchase_amount”.
- Engineered a new age_group feature by splitting customer age into four equal-sized quartile bins: Young Adult, Adult, Middle-Aged, and Senior.

- Engineered a `purchase_frequency_days` feature, converting categorical frequency labels (e.g. “Weekly”, “Fortnightly”, “Monthly”) into a numeric day-count for quantitative analysis.
- Verified that the `promo_code_used` and `discount_applied` columns were fully duplicative (identical in every row) and removed the redundant `promo_code_used` column.
- Loaded the cleaned dataset into a PostgreSQL database via SQLAlchemy for structured querying.

3.2 Business Analysis - SQL (PostgreSQL)

Ten targeted business questions were answered directly against the cleaned dataset using PostgreSQL, covering revenue analysis, customer segmentation, product performance, and behavioural comparisons. Each query and its result is presented in full in Section 5 of this report.

3.3 Visualisation - Microsoft Power BI

An interactive Power BI dashboard (`Customer_Behaviour_Dashboard.pbix`) was built on top of the cleaned dataset to allow ongoing, self-service exploration of the same metrics covered in the SQL analysis - revenue by category and gender, top-rated and most-discounted products, customer segment distribution, and age-group revenue contribution - with interactive filtering by category, gender, season, and location.

3.4 Technology Stack Summary

Layer	Tool	Purpose
Data Cleaning	Python 3 / pandas / Jupyter	Null handling, feature engineering, schema standardisation
Data Storage	PostgreSQL / SQLAlchemy	Structured storage for SQL-based business querying
Business Querying	SQL (PostgreSQL dialect)	Ten targeted business questions (aggregation, window functions, CTEs)
Visualisation	Microsoft Power BI	Interactive dashboard for ongoing self-service analysis
Reporting	Python (pandas / matplotlib)	Chart generation and statistical summarisation for this report

4 Dataset Overview

The cleaned dataset contains 3,900 unique customer transactions with no duplicate customer IDs. After cleaning, the dataset has zero missing values in any analytically relevant field. The table below summarises the core structural characteristics of the dataset.

Metric	Value
Total transaction records	3,900

Metric	Value
Attributes per record	18 (post-cleaning: 19, incl. engineered fields)
Unique customers	3,900 (one transaction per customer)
Unique products	25
Product categories	4 (Clothing, Accessories, Footwear, Outerwear)
Customer locations (U.S. states)	50
Age range	18 – 70 years
Purchase amount range	\$20 – \$100
Payment methods	6 (Credit Card, PayPal, Cash, Debit Card, Venmo, Bank Transfer)
Missing values after cleaning	0

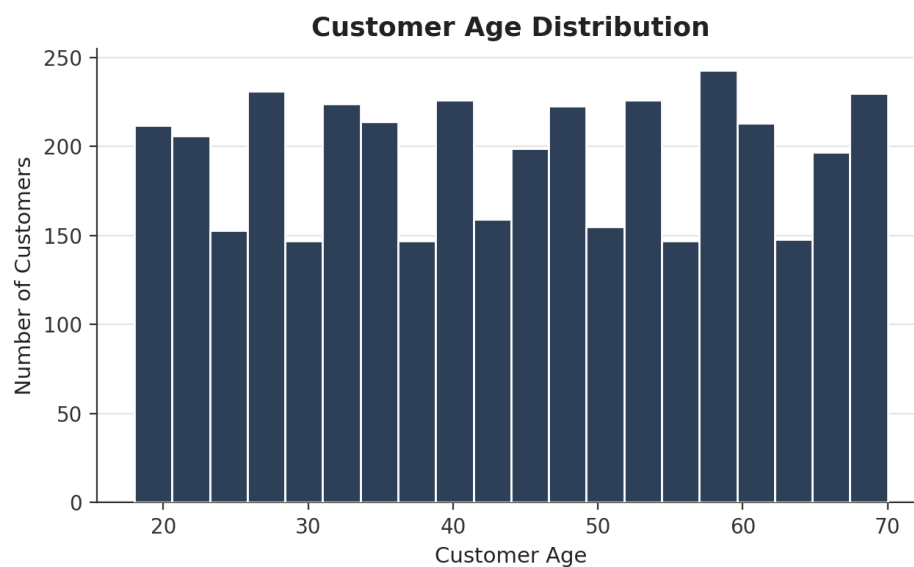


Figure 4.1 - Distribution of customer age across the dataset (18–70 years).

5 Detailed Business Question Analysis

This section presents each of the ten core business questions, the exact SQL query used to answer it, the resulting output, a supporting visualisation, and the business insight it reveals.

5.1 Revenue Contribution by Gender

SQL Query

```
SELECT gender, SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

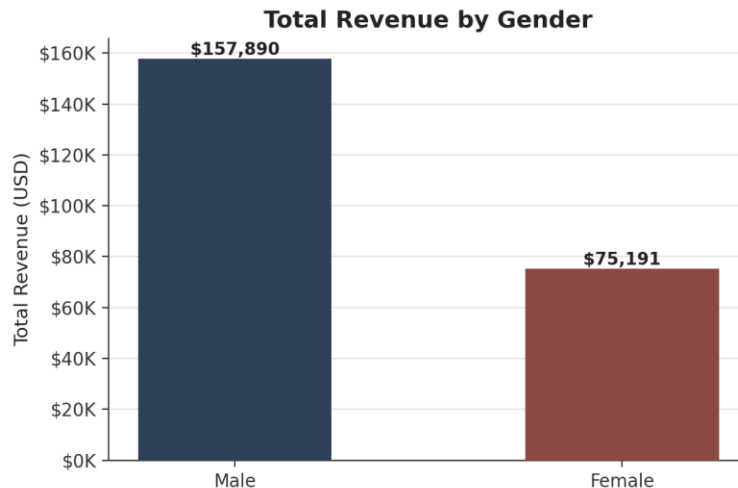


Figure 5.1 - Total revenue generated by male vs. female customers.

Insight

- Male customers account for 68% of all transactions and 67.7% of total revenue (\$157,890 vs. \$75,191).
- This is primarily a volume effect - the average order value is broadly similar across genders - so the revenue gap is driven by customer mix, not by male customers spending more per transaction.
- Marketing and retention budget could be rebalanced to test whether targeted campaigns grow the underrepresented female customer base.

5.2 High-Spending Customers Who Used a Discount

SQL Query

```
SELECT customer_id, purchase_amount AS more_spent_customers
FROM customer
WHERE discount_applied = 'Yes'
AND purchase_amount >= (SELECT AVG(purchase_amount) FROM customer);
```

Insight

- 839 customers (21.5% of the full customer base) used a discount and still spent at or above the overall average purchase amount of \$59.76.
- This indicates a meaningful segment of price-conscious-but-high-value shoppers - discounts are not purely a race-to-the-bottom lever; they can coexist with strong order values.

- This segment is a strong candidate for tiered loyalty offers, since they already demonstrate willingness to spend above average even when incentivised.

5.3 Top 5 Products by Average Review Rating

SQL Query

```
SELECT item_purchased, ROUND(AVG(review_rating::NUMERIC),2) AS top_5_products
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC LIMIT 5;
```

Result

Rank	Product	Avg. Rating
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.79

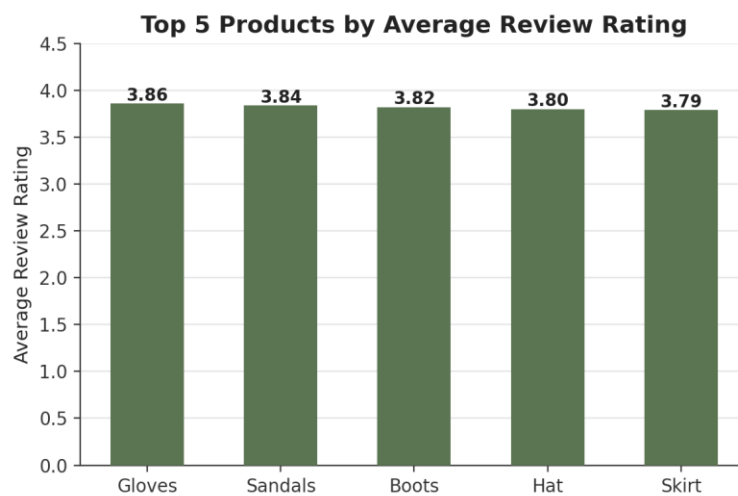


Figure 5.2 - Top five products ranked by average customer review rating.

Insight

- Gloves (3.86), Sandals (3.84) and Boots (3.82) are the highest-rated products in the catalogue, all from the Footwear and Accessories categories.
- The narrow spread (3.79–3.86) suggests overall customer satisfaction is fairly consistent across top products rather than being driven by one standout item.
- These top-rated products are strong candidates for feature placement in marketing creative and homepage merchandising.

5.4 Average Spend: Standard vs. Express Shipping

SQL Query

```
SELECT shipping_type, ROUND(AVG(purchase_amount),2) AS avg_purchase_amount
FROM customer
WHERE shipping_type in ('Standard','Express')
GROUP BY shipping_type;
```

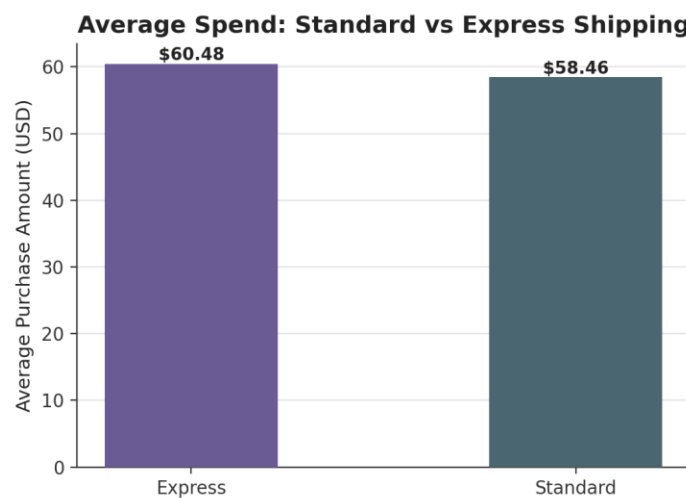


Figure 5.3 - Average purchase amount by shipping method.

Insight

- Customers who select Express shipping spend \$60.48 on average, slightly higher than the \$58.46 average for Standard shipping - a difference of about \$2.02 (3.5%).
- The gap is small enough that shipping preference alone is not a strong predictor of order value, though it may correlate with urgency-driven or gift purchases.

5.5 Subscriber vs. Non-Subscriber Spend Comparison

SQL Query

```
SELECT subscription_status, COUNT(customer_id) AS total_customers,
ROUND(AVG(purchase_amount),2) AS avg_spend,
ROUND(SUM(purchase_amount),2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY avg_spend, total_revenue DESC;
```

Result

Subscription Status	Customers	Avg. Spend	Total Revenue
Yes	1,053	\$59.49	\$62,645
No	2,847	\$59.87	\$170,436

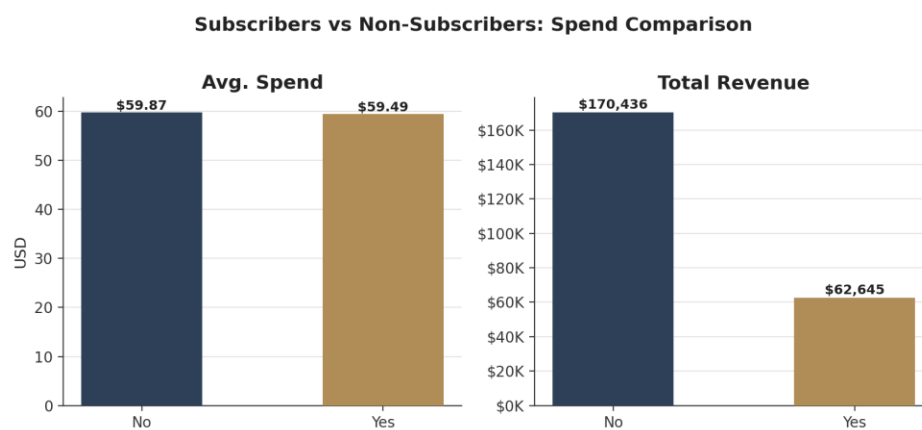


Figure 5.4 - Customer count, average spend and total revenue by subscription status.

Insight

- Only 27.0% of customers (1,053 of 3,900) are subscribers, yet subscribers do not out-spend non-subscribers on a per-order basis: \$59.49 average vs. \$59.87 for non-subscribers.
- Because this dataset captures one transaction per customer, subscription status currently shows no measurable uplift in single-order value - its value proposition likely lies in repeat-purchase frequency rather than basket size, which this snapshot cannot fully test.
- Recommend tracking subscriber lifetime value over multiple orders rather than judging the programme on per-transaction spend alone.

5.6 Top 5 Products by Discount Usage Rate

SQL Query

```
SELECT item_purchased,
```

```
ROUND(100 * SUM(CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END)/COUNT(*),2) AS
discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC LIMIT 5;
```

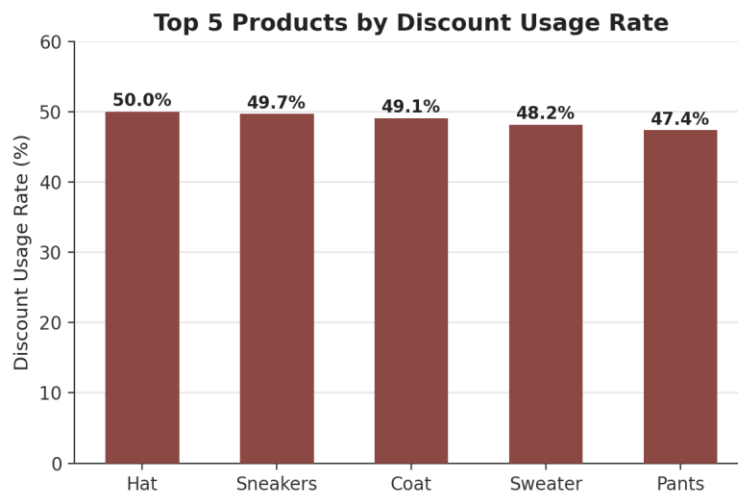


Figure 5.5 - Products with the highest share of discount-applied purchases.

Insight

- Hats have the highest discount usage rate at 50%, followed closely by Sneakers (49.7%), Coats (49.1%), Sweaters (48.2%) and Pants (47.4%).
- These products may be more price-sensitive or more frequently included in promotional campaigns; worth investigating whether high discount reliance is eroding their margin contribution relative to full-price sellers.

5.7 Customer Segmentation by Purchase History

SQL Query

```
WITH customer_type AS (
  SELECT customer_id, previous_purchases,
  CASE
    WHEN previous_purchases = 1 THEN 'New'
    WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
    ELSE 'Loyal'
  END AS customer_segment
  FROM customer
)
SELECT customer_segment, COUNT(*) AS number_of_customers
FROM customer_type
GROUP BY customer_segment;
```

Result

Segment	Definition	Customers	% of Total
Loyal	> 10 previous purchases	3,116	79.9%
Returning	2–10 previous purchases	701	18.0%
New	1 previous purchase	83	2.1%

Customer Segmentation by Purchase History

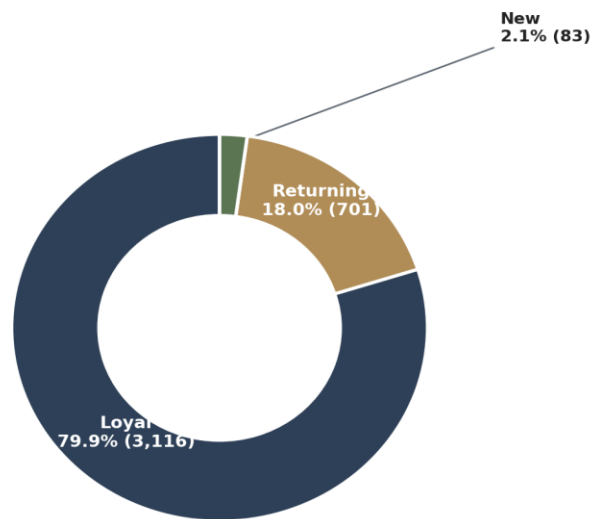


Figure 5.6 - Customer distribution across New, Returning and Loyal segments.

Insight

- The customer base is overwhelmingly Loyal (3,116 customers, 79.9%), with far smaller Returning (701, 18.0%) and New (83, 2.1%) segments.
- This heavy skew toward loyal, high-frequency purchasers suggests the sampled population is largely composed of established shoppers rather than recent acquisitions - new customer acquisition may be an area needing more attention relative to retention.

5.8 Top 3 Best-Selling Products per Category

SQL Query

```
WITH item_counts AS (
```

```

SELECT category, item_purchased,
       COUNT(customer_id) AS total_orders,
       ROW_NUMBER() OVER (PARTITION BY category ORDER BY COUNT(customer_id) DESC) AS item_rank
FROM customer
GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;

```

Result

Category	Rank 1	Rank 2	Rank 3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	-

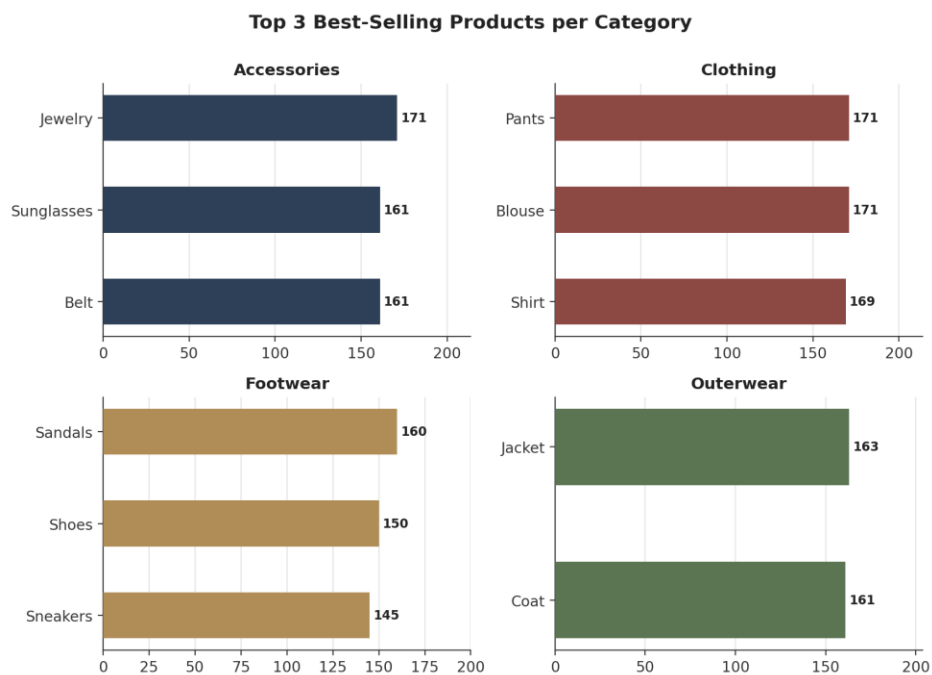


Figure 5.7 - Top three best-selling products within each product category.

Insight

- Jewelry (171 orders) leads Accessories; Blouse (171) leads Clothing; Sandals (160) leads Footwear; and Jacket (163) leads Outerwear.

- Order counts for the top 3 products within each category are tightly clustered, indicating balanced demand across a product's category rather than a single dominant “hero” SKU - useful for inventory planning across the full assortment rather than concentrating stock in one item.

5.9 Subscription Propensity Among Repeat Buyers

SQL Query

```
SELECT subscription_status,  
COUNT(customer_id) AS repeat_buyers  
FROM customer  
WHERE previous_purchases > 5  
GROUP BY subscription_status;
```

Subscription Status of Repeat Buyers (>5 Prior Purchases)

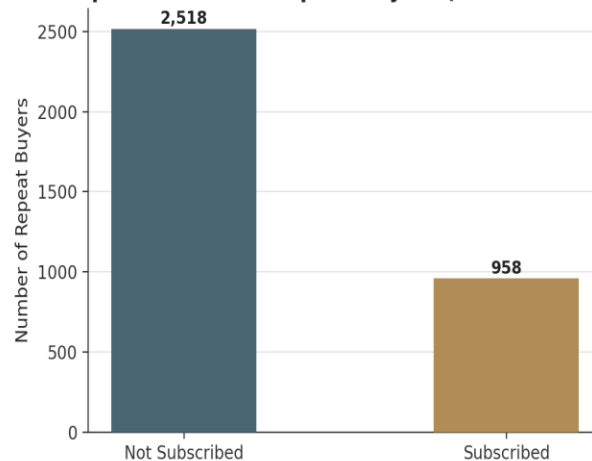


Figure 5.8 - Subscription status among customers with more than five previous purchases.

Insight

- Among the 3,476 customers with more than five previous purchases, only 27.6% (958) are subscribed - almost identical to the 27.0% subscription rate across the full customer base.
- This indicates that being a frequent, repeat buyer does not currently correlate with a higher likelihood of subscribing, suggesting an untapped opportunity: repeat buyers who are not yet subscribers are a natural, low-friction target audience for subscription conversion campaigns.

5.10 Revenue Contribution by Age Group

SQL Query

```
SELECT age_group,
```

```
SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Result

Age Group	Total Revenue
Young Adult	\$62,143
Middle-Aged	\$59,197
Adult	\$55,978
Senior	\$55,763

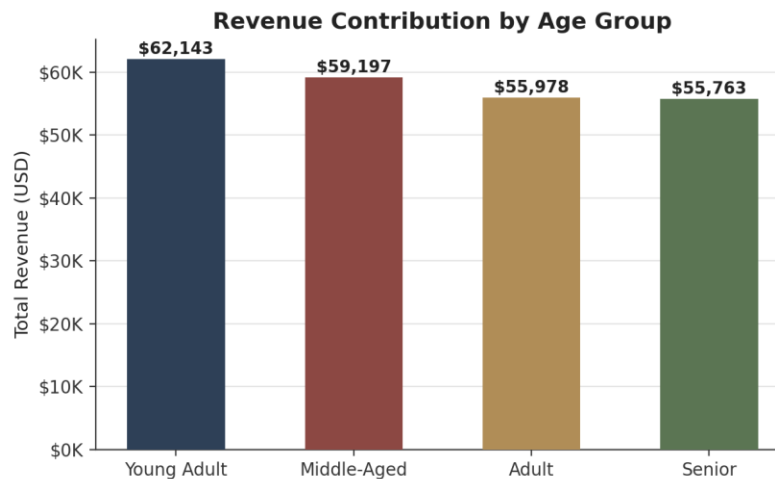


Figure 5.9 - Total revenue generated by each engineered age group (quartile-based).

Insight

- Revenue is remarkably evenly spread across all four age quartiles: Young Adult (\$62,143), Middle-Aged (\$59,197), Adult (\$55,978) and Senior (\$55,763) - a spread of only about 10% between the highest and lowest group.
- No single age group dominates revenue, meaning age-targeted marketing should be used to fine-tune messaging and product mix rather than to concentrate budget on one demographic.

6 Supplementary Insights

Beyond the ten core business questions, several additional cuts of the data provide useful context for merchandising and channel strategy.

6.1 Revenue by Product Category

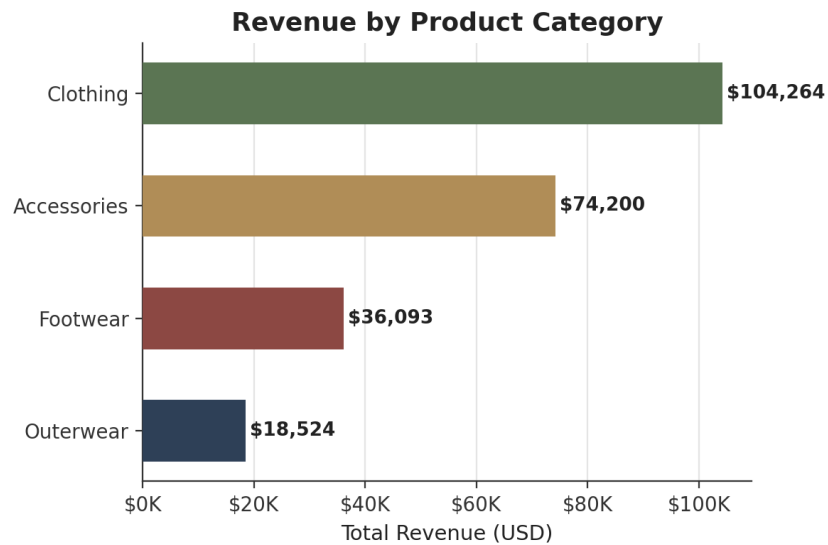


Figure 6.1 - Total revenue by product category.

Clothing dominates category revenue at \$104,264 (44.7% of total), more than the combined revenue of Footwear and Outerwear. Accessories is a strong second at \$74,200 (31.8%), while Footwear (\$36,093) and Outerwear (\$18,524) are comparatively niche in this dataset.

6.2 Revenue by Season

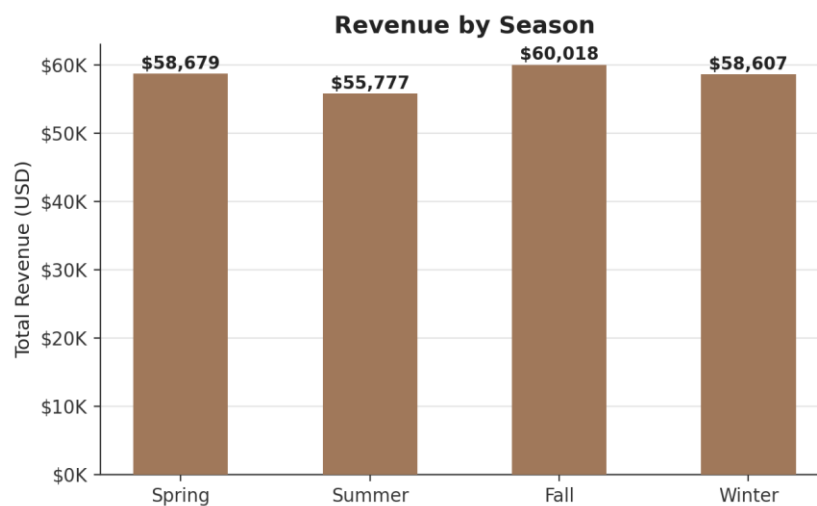


Figure 6.2 - Total revenue by season of purchase.

Revenue is well balanced across the four seasons, ranging narrowly from \$55,777 (Summer) to \$60,018 (Fall) - a spread of only 7.6%. This suggests the product catalogue is not strongly seasonal and demand is sustained year-round, which is favourable for inventory and cash-flow planning.

6.3 Revenue by Payment Method

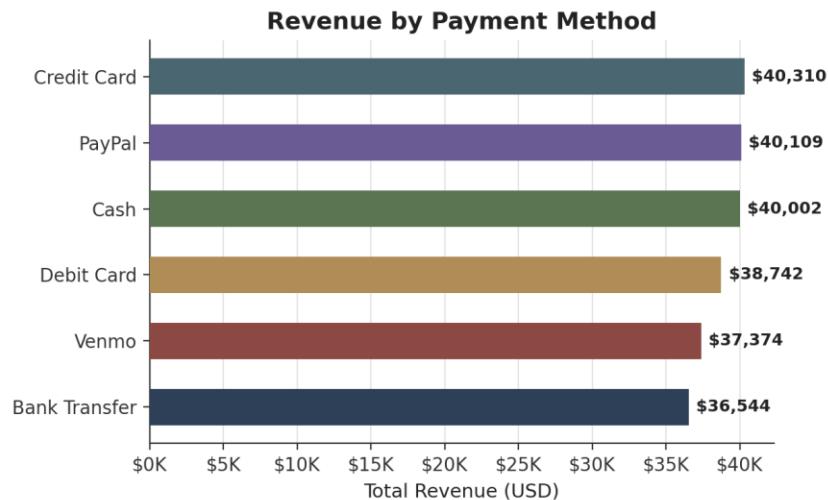


Figure 6.3 - Total revenue by payment method.

Payment method usage is highly balanced, with Credit Card (\$40,310) and PayPal (\$40,109) leading marginally over Cash, Debit Card, Venmo and Bank Transfer. No single payment channel dominates, indicating checkout flexibility is being well utilised by the current customer base and no payment method should be deprioritised.

7 Interactive Power BI Dashboard

In addition to the static SQL-driven analysis presented above, an interactive Power BI dashboard (Customer_Behaviour_Dashboard.pbix) was developed on the same cleaned dataset. The dashboard mirrors the core metrics covered in this report - revenue by gender and category, top-rated and most-discounted products, customer segmentation, and age-group revenue - but allows business stakeholders to interactively filter and drill down by category, gender, season, location and subscription status without requiring SQL knowledge.

Dashboard Capabilities

- Dynamic slicers for gender, category, season, location and payment method.
- Cross-filtering visuals so that selecting one chart element (e.g. a category) instantly updates all other visuals on the page.
- KPI cards summarising total revenue, average order value, transaction count and average review rating in real time as filters are applied.
- A reusable, refreshable data model - as new transactions are appended to the source data, the dashboard can be refreshed to reflect current performance without rebuilding any visuals.

The .pbix file is included alongside this report as a supporting deliverable and is intended for use by business and merchandising stakeholders for ongoing, self-service monitoring beyond the point-in-time snapshot captured in this document.

8 Conclusions & Recommendations

8.1 Summary of Conclusions

- The customer base is heavily loyalty-driven (79.9% “Loyal” segment) and male-skewed (68% of transactions), with Clothing as the clear revenue-leading category.
- Discounting is widespread (43% of transactions) but does not uniformly suppress spend - a meaningful share of discount users still spend above average.
- Subscription status, shipping type, and age group each show only modest differences in per-order spend, suggesting that basket value is fairly consistent across customer segments and that differentiation is more likely to emerge in purchase frequency and lifetime value than in single-order size.
- Demand is well balanced across seasons and payment methods, indicating a stable, non-seasonal revenue base with flexible checkout adoption.

8.2 Recommendations

- Investigate the gender revenue gap: pilot targeted campaigns to grow female customer acquisition and basket size, given the current 68/32 transaction split.
- Convert high-frequency non-subscribers: repeat buyers (>5 previous purchases) subscribe at the same 27% rate as the general population - target this group specifically with subscription incentives, since they have already demonstrated loyalty.
- Review discount strategy on high-discount-rate items (Hat, Sneakers, Coat, Sweater, Pants) to confirm promotions are driving incremental volume rather than eroding margin on products that would sell regardless.
- Prioritise New-segment acquisition: at only 2.1% of the customer base, new-customer growth appears to be a smaller focus than retention - consider rebalancing acquisition spend to sustain the loyal base long-term.
- Leverage top-rated products (Gloves, Sandals, Boots, Hat, Skirt) in featured placements and marketing creative to capitalise on demonstrated customer satisfaction.
- Adopt the Power BI dashboard for ongoing monitoring so that merchandising and marketing teams can self-serve updated metrics between formal reporting cycles.

8.3 Suggested Next Steps

- Extend the dataset to multiple transactions per customer to properly measure subscriber lifetime value and repeat-purchase frequency rather than single-order spend.
- Layer in cohort analysis to track how the New → Returning → Loyal segment pipeline evolves over time.
- Add profitability (cost/margin) data to complement the current revenue-only view, enabling true ROI assessment of discount and shipping strategies.